

Assessment of Groundwater Quality Using Water Quality Index (WQI), Pollution Index of Groundwater (PIG), and GIS in Parts of Anantapur District, Andhra Pradesh, India

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Abstract

The purpose of this study was to assess the potability of groundwater resources in some parts of the Anantapur district of the Indian state of Andhra Pradesh. The water quality index, a spatial evaluation of groundwater using a GIS-based interpolation method, and the polluting groundwater index are some of the methods used to analyze water quality and explain the water quality of the studied area. A total of 22 groundwater samples were obtained during the post-monsoon period. The samples were tested for pH, EC, TDS, TH, and the major cations and anions. PIG levels between 0.98 and 1.98, with an average of 1.36, categorize low contamination in 5% of the research area and groundwater that is fit for human consumption. Additionally, 77% of the groundwater samples have drinking qualities that are only slightly acceptable. WQI values ranged from 85.97 to 113.52 mg/L, indicating that 55% and 45% of the samples are safe for drinking, respectively, while the other 45% are not. Groundwater samples are found in the field of rock dominance, according to the Gibbs plot. GIS techniques are being used to analyze the spatial variation of groundwater quality, and the results show that the majority of groundwater samples marginally meet standards for potable water, necessitating prior treatment before use.

Keywords: Groundwater, Hydro geochemistry, PIG, WQI, GIS, Spatial distribution Maps.

INTRODUCTION

Water is necessary for human survival, as well as all forms of development. Huge population expansion, increased urbanization, expanding industries, and massive agricultural operations around the world have all contributed to a massive increase in the mandate for freshwater for home, agricultural, and industrial uses in recent years [1–2]. Groundwater is a life elixir, but it has been plagued by challenges in recent years, particularly in terms of quality, particularly in dry and semiarid areas. Agricultural pollutants, such as chemicals, insecticides, and other metal contamination, make water unsafe for human consumption. As a result, understanding the hydrogeochemistry of water is

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critical for evaluating water for drinking, agriculture, and other uses. Understanding groundwater occurrence, replenishment, and recovery is especially critical in dry and semi-arid areas because of disparities in monsoonal precipitation, lacking shallow waters, and over-drafting of groundwater resources. Aquifer quality is affected by recharged water, meteorological rainfall, inland groundwater, and underground geochemical processes. As a result, knowing hydrochemistry is crucial for evaluating groundwater quality in any location where groundwater is used for drinking [3–4]. Water excellence testing can reveal important evidence